



Customer Shopping Behavior Analysis

Objective: use 3,900 transactions to reveal spending patterns, segments, preferences, subscriptions.



Dataset Summary

Scope

Rows: 3,900 · Columns: 18

Key Features

Demographics, purchase details, behavior, subscription, shipping

Missing Data

37 missing values in Review Rating

EDA & Preparation (Python)

01

Load & Inspect

pandas: df.info(), .describe()

02

Missing Values

Imputed Review Rating by category median

03

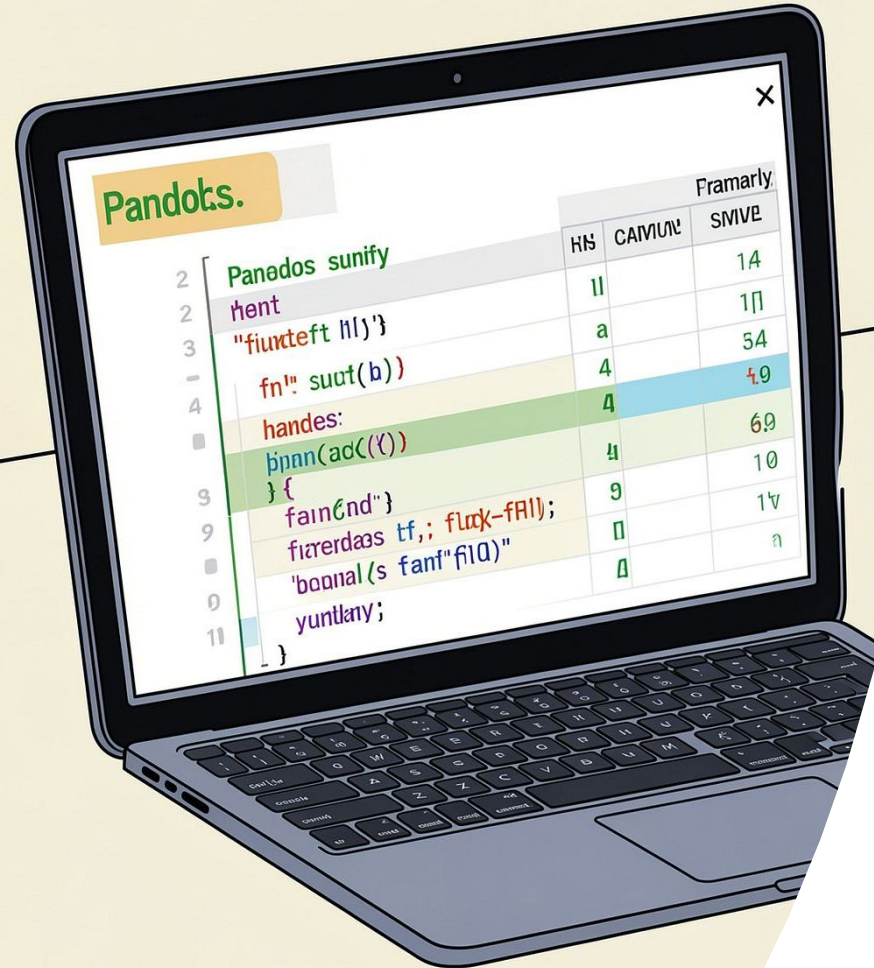
Standardize

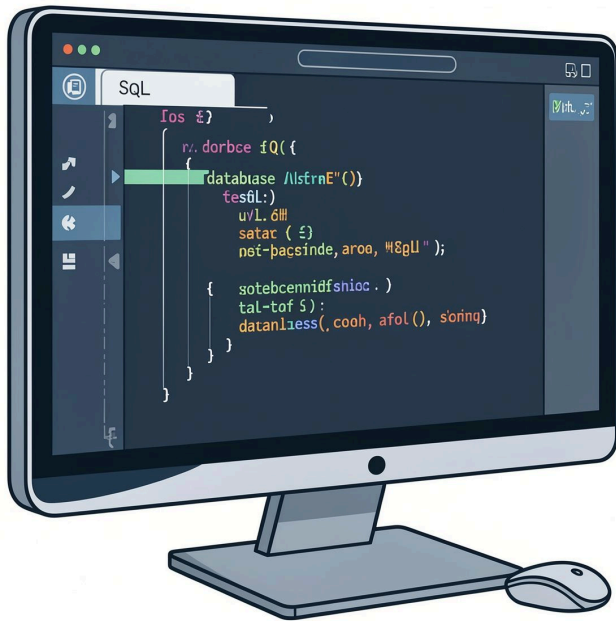
Renamed columns to snake_case

04

Feature Eng.

age_group, purchase_frequency_days





SQL Analysis (MYSQL)

- Loaded cleaned DataFrame to DB
- Ran business queries for revenue, segments, product metrics

NAL FMNALE S REVENUE



Revenue & Demographics



By Gender

Male revenue: 157,890 · Female revenue: 75,191



By Age Group

Young Adult 62,143 · Middle-aged 59,197 · Adult 55,978 · Senior 55,763

Discounts & Product Insights



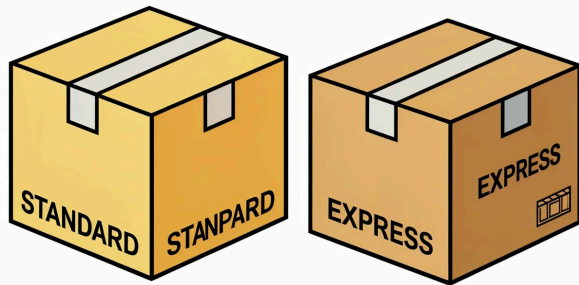
Discount-Dependent Products

Top: Hat 50% · Sneakers 49.66% · Coat 49.07% ·
Sweater 48.17% · Pants 47.37%



Top Rated Items

Gloves 3.86 · Sandals 3.84 · Boots 3.82 · Hat
3.80 · Skirt 3.78



Operations & Subscriptions

Shipping Comparison

Standard avg: 58.46 ·
Express avg: 60.48

Subscribers

Yes: 1,053 customers
· Avg spend 59.49 ·
Revenue 62,645

Non-Subscribers

No: 2,847 customers
· Avg spend 59.87 ·
Revenue 170,436



LOYAL



RETURNING



RETURNING



SNEW



CUSTOMER



ROYAL



NEW



Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

New

83 customers



Power BI Dashboard

Interactive view: 3.9K customers · Avg purchase \$59.76 · Avg rating 3.75

- Subscription split: Yes 27% · No 73%
- Revenue by category: Clothing 100K · Accessories 75K · Footwear 30K · Outerwear 15K
- Sales by category: Clothing 1,800 · Accessories 1,200 · Footwear 600 · Outerwear 300

Key Insights & Recommendations

Insights

High male revenue; loyal segment dominant; discount-dependency on specific products

Recommendations

- Boost subscriptions with exclusive benefits
- Implement loyalty rewards for repeat buyers
- Review discount policy to protect margins
- Promote top-rated & best-selling items
- Target marketing to high-revenue age groups & express-shipping users

Conclusion

Data-driven steps to increase revenue, retention, and margin control.

